

ABIJITH BINU

AI/ML ENGINEER

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SUMMARY

Aspiring AI/ML Engineer and MSc Artificial Intelligence student with a strong foundation in Python, Deep Learning, and Computer Vision. Hands-on experience building Generative AI applications, REST APIs, and deploying local LLMs. Eager to apply academic learning and practical project skills to innovative AI-driven software development teams.

PROFESSIONAL EXPERIENCE

AI/ML Trainee (Internship Program), Linnk Academy India

Oct 2025 – Present

- Undergoing intensive training in Machine Learning algorithms, Neural Networks, and MLOps practices.
- Building and deploying REST APIs using FastAPI and Flask to serve ML models as scalable web services.
- Focusing on structured data analysis, feature engineering, and model evaluation techniques.

Graduate Researcher, Mahatma Gandhi University

Jan 2026 – Present

- Researching lightweight model deployment on edge devices for inclusive education (Edge-AI project).
- Investigating quantization and pruning techniques to optimize inference latency on low-power hardware.
- Exploring integration of local LLMs via Ollama for offline, privacy-preserving AI on edge devices.

EDUCATION

Master of Science (MSc) in Artificial Intelligence

2025 – Present

Mahatma Gandhi University (Cochin Arts & Science College)

Bachelor of Computer Applications (BCA)

2022 – 2025

Mahatma Gandhi University (Cochin Arts & Science College)

PROJECTS

SignBridge — ASL-to-English Translation TensorFlow LSTM, MediaPipe, Flask, React, Ollama, Mistral

- Built a real-time offline ASL-to-English translation system for accessible communication.
- Trained a custom LSTM model on 9,200 samples across 36 sign classes.
- Integrated a locally hosted Mistral model via Ollama to generate natural English output.
- Added text-to-speech for real-time spoken communication.

Pixelarc — AI Image Upscaler React Native , Flask , Real-ESRGAN , Expo , Render

- Built a full-stack mobile application that upscales images using Real-ESRGAN AI inference.
- Developed a Flask backend integrating the ncnn-Vulkan binary for GPU-accelerated upscaling on local hardware, with a cloud deployment on Render using Pillow fallback for free-tier compatibility

Multimodal Video Summarizer Python, Whisper, Llama 3, BLIP, Streamlit

- Developed a Generative AI tool using Whisper for audio transcription and Llama 3 for text summarization.
- Integrated BLIP for visual image captioning, enabling multimodal content understanding from video inputs.

Local LLM Deployment with Ollama Python, Ollama, Llama 3

- Set up Ollama to run LLMs locally and integrated offline inference into Python apps with full data privacy.

ML Model API Service Python, FastAPI, Flask, Docker

- Designed RESTful APIs using FastAPI and Flask to expose trained ML models as production-ready endpoints.
- Containerized services with Docker for consistent, portable deployment across environments.

TECHNICAL SKILLS

- Languages & Core: Python, C++, Java, SQL, Data Structures & Algorithms
- AI/ML Frameworks: TensorFlow, PyTorch, Scikit-learn, OpenCV, MediaPipe
- Generative AI & NLP: LLMs (Llama 3), Transformers, Whisper, BLIP, Hugging Face
- API Development: FastAPI, Flask, RESTful API Design
- Deployment & Tools: Docker, Streamlit, Git, Linux, MLOps Basics, Ollama
- Languages: English (Fluent), Malayalam (Fluent)